Damiano Franzò | CV

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danzio • **DoB** : 03/05/1995

Msc student currently at second year of Double Degree program of Computer Engineer, Data science track, at Politecnico di Torino, and Data Science and Engineering at EURECOM. Passionate about machine learning, with proven experience in C++, Python and PyTorch. Oriented toward challenges that require strong problem solving skills and a R&D approach. Always improving soft skills and teamwork attitude. Looking for a 6-months internship starting from July 2019.

Experience

Work Experience

- **Student Researcher** — **Torino**
  ERC 321 Action — *May 2018 - October 2018*
  Research studentship "Ad hoc computational frameworks in numerical eletromagnetics". Implemented computational electromagnetics algorithms in C++ within the framework. Acquired practical experience of Linear Algebra, Computational Electromagnetics and git.
- **Tutor** — **Torino**
  Tutoring for students — *September 2014 – September 2018*
  Provided private lessons for high school students regarding Maths, physics and coding

Projects

- **Performance and Convergence Analysis of Asynchronous Distributed Machine Learning with PyToch**
  Project under development for EURECOM
  Research project with the supervision of Prof. Michiardi. Analsysis of Asyncronous and Syncronous distributed machine learning models. The goal is to to determine how the staleness of distributed asynchronous deep learning models influences the convergence and the performance of the whole system.
- **Triangulation and analysis of mobile devices: ESP tracking system**
  Developed a distributed system to gather, analyze and store data related to triangulation and analysis of mobile devices in buildings. Measuring devices signals power through chip micro-controllers placed around rooms.
- **Analysis of ISR delays on a Raspberry Pi 3**
  Developed kernel modules that run an ISR. Performed experiments to understand how the load of work of the Raspberry influences the delay of the output square wave.

Education

Academic Qualifications

- **EURECOM** — **Biot**
  Data science and Engineering, — *Oct 2018–Now*
- **Politecnco di Torino** — **Torino**
  Computer Engineering, Data science track, Average Grade 29.0/30 (top 3%) — *Oct 2017–Now*
- **Politecnico di Torino** — **Torino**
  Computer Engineering, Finale Grade 105/110 — *2014–2017*
- **Liceo Superiore O.M. Corbino** — **Siracusa**
  Scientific high school in Sicily — *2009–2014*

Student volunteering experience

- **Eta Kappa Nu - Polito** — **Torino**
  Head of HR — *October 2017– October 2018*
  Performed interviews during recruiting sessions. Organized events for team building among members.

Extracurricular activities

- **ESoA course 2018: "Advanced Computational Electromagnetics"** *course addressing frequency-domain, integral-equation based computational techniques for the analysis of challenging electromagnetic problems. The challenges come from the need to apply computational electromagnetics to real-life antenna and antenna platform design.*
- **Deep Learning Prerequisite courses - Udemy** *Attended several online courses thought Udemy platform*
  Deep Learning in Python
  Deep Learning Prerequisite: Linear Regression in Python
  Deep Learning Prerequisite: Logistic Regression in Python
  Deep Learning Prerequisites: The Numpy Stack in Python
- **Summer School Alfaclass 2015** *Attended a Summer School of Mathematics of University of Turin in collaboration with Polytechnic of Turin, an opportunity given only to 30 students of Polytechnic of Turin.*

Technical and Personal skills

Technical skills

Domain knowledge on:
Computational Electromagnetics
Machine learning
Deep Learning
Convex optimization
Cryptography
Operating systems
Multithreaded Programming
Big Data: MapReduce, Hadoop, Spark

Programming Languages

- **Low Level:** C, ARM assembly
- **Object Oriented:** C++, C#, Python, Java
- **Specific Domain:** Matlab, Julia, SQL
- **Web Programming:** HTML, CSS, JS, PHP
- **Scripting:** AWK, Bash
- **Libraries:** NumPy, TensorFlow, PyTorch, Keras, Scikit-learn, Pandas

Personal Skills

- **Tools:** Git, Docker, Slack, G suite
- **Other:** Touch typing racer (speed close to 80 WPM), Mnemonic techniques knowledge

Languages

- **Italian:** Mother-tongue
- **English:** C1
- **French** B1
- **German** A1.2

Awards

- **Italian olympics of astronomy 2013:** Finalist at national olympics of astronomy.

References

Research studentship for ERC 321 Action:
Prof. Andriulli Francesco
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